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MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 7.2

Quadratic Forms

Quadratic Forms

A quadratic form on $\mathbb{R}^n$ is a function Q defined on $\mathbb{R}^n$ whose value at a vector $\mathbf{x}$ can be computed by an expression of the form

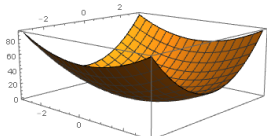
$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

where A is an $n \times n$ symmetric matrix.

A is called the matrix of the quadratic form.

Example 1

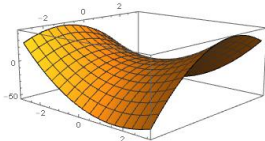
Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Compute $\mathbf{x}^T A \mathbf{x}$ for the following matrix.

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 7 \end{bmatrix}$$


Example 2

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Compute $\mathbf{x}^T A \mathbf{x}$ for the following matrix.

$$A = \begin{bmatrix} 4 & 2 \\ 2 & -5 \end{bmatrix}$$



Example 3

Find the matrix of the quadratic form.

$$3x_1^2 - 4x_1x_2 + 5x_2^2$$

Example 4

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Compute $\mathbf{x}^T A \mathbf{x}$ for the following matrix.

$$A = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

Example 5

Find the matrix of the quadratic form.

$$5x_1^2 + 3x_2^2 + 2x_3^2 - x_1x_2 + 8x_2x_3$$

Section 4.4:

Change of Coordinates Matrices

To change coordinates from a basis B to the standard basis in $\mathbb{R}^n$, we can compute

$$\mathbf{x} = P_B[\mathbf{x}]_B$$

for some matrix P_B . In general, we call P_B the change of coordinates matrix from B to the standard basis in $\mathbb{R}^n$.

Similarly, we can go the other direction using the inverse of P_B .

$$P_B^{-1}\mathbf{x} = [\mathbf{x}]_B$$

Change of Variables with Quadratic Forms

Simplified notation:

$$\text{Let } \mathbf{x} = P\mathbf{y} \text{ (or, equivalently, } \mathbf{y} = P^{-1}\mathbf{x})$$

If we apply such a change to $\mathbf{x}^T A \mathbf{x}$, we get

$$\mathbf{x}^T A \mathbf{x} = (P\mathbf{y})^T A (P\mathbf{y}) = \mathbf{y}^T P^T A P \mathbf{y} = \mathbf{y}^T (P^T A P) \mathbf{y}$$

Since A is symmetric, we can pick an orthogonal matrix P such that $P^T A P = D$ for a diagonal matrix D .

If we pick such a matrix P , then

$$\mathbf{x}^T A \mathbf{x} = \mathbf{y}^T D \mathbf{y}$$

Example 6

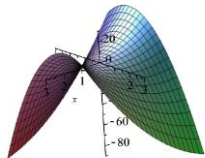
Make a change of variable $\mathbf{x} = P\mathbf{y}$ that transforms the quadratic form

$$2x_1^2 + 6x_1x_2 - 6x_2^2$$

into one with no cross-product term. Write the new quadratic form

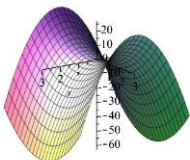
Example 6 (continued)

Original matrix: $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$
 $z = Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$



Example 6 (continued)

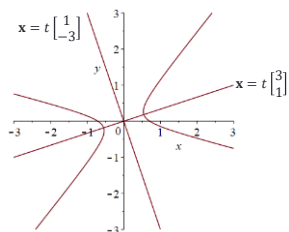
New matrix: $D = \begin{bmatrix} -7 & 0 \\ 0 & 3 \end{bmatrix}$
 $z = Q(\mathbf{y}) = \mathbf{y}^T D \mathbf{y}$



Example 6 (continued)

$$z = 2x^2 + 6xy - 6y^2$$

For clarity, pick $z = 1$



The Principal Axes Theorem

Let A be an $n \times n$ symmetric matrix. Then, there is an orthogonal change of variable $\mathbf{x} = P\mathbf{y}$ which transforms the quadratic form $\mathbf{x}^T A \mathbf{x}$ into a quadratic form $\mathbf{y}^T D \mathbf{y}$ with no cross-product term.

The columns of P are the principal axes of the quadratic form $\mathbf{x}^T A \mathbf{x}$.
